PRELIMINARY DESIGN REVIEW (PDR++)
WING STRUCTURAL DESIGN DOCUMENT

Nomenclature

W = Aircraft weight (N)
n = Load factor
L = Lift force (N)
b = Wingspan (m)
S = Wing area (m^2)
c = Mean aerodynamic chord (m)
w(y) = Spanwise load distribution (N/m)
V(y) = Shear force (N)
M(y) = Bending moment (Nm)
T = Torsional moment (Nm)
e = Moment arm (m)
EI = Bending stiffness (Nm^2)
GJ = Torsional stiffness (Nm^2)
A66 = Laminate shear stiffness (N/m)
A_m = Enclosed area (m^2)
delta_tip = Wing tip deflection (m)

Aircraft Parameters

MTOW = 580 kg
W = 5680 N
b = 11 m
Semi-span = 5.5 m
S = 13.5 m^2
Aspect Ratio = 9
c = 1.23 m
Vc = 50 m/s
Vs = 23 m/s

Airfoil Definition

Airfoil: NACA 2412
Thickness ratio = 0.12
Maximum thickness = 0.15 m
CL_max ≈ 1.5
Re = 2e6 to 4e6

Load Factors

Limit: n = +3.8, -1.52
Ultimate: n_u = 1.5*n

Lift Calculation

L = n * W
Positive: 3.8 * 5680 = 21584 N
Per wing = 10792 N

Spanwise Load Distribution

w(y) = w0 * sqrt(1 - (2*y/b)^2)
w0 = 4*L / (pi*b) ≈ 1250 N/m

Shear Force

V(y) = integral of w(y) dy
Root shear ≈ 10800 N

Bending Moment

M_root = (L * b) / (3*pi) ≈ 20000 Nm (limit)
Ultimate ≈ 30000 Nm

Torsional Loads

T = L * e
e = 0.1*c ≈ 0.123 m
T ≈ 1300 Nm

Additional Load Cases

Gust loads: delta_n ≈ 1.5
Fuel load ≈ 300 N
Landing loads
Control surface loads

Structural Model

Closed wing box with front spar, rear spar, skins, ribs. Load carried by spar caps (bending), webs (shear), skins (torsion).

Sectional Properties

EA = axial stiffness
EI = bending stiffness
GJ = torsional stiffness

Torsion Equation

GJ = 4*A_m^2 / integral(ds / A66)

Deflection Requirement

delta_tip <= 0.4 m at limit load
Required EI >= 4.5e6 Nm^2

Failure Modes

Fiber failure
Matrix cracking
Buckling (skin/web/cap)
Delamination
Shear failure

Deliverables

Wing box layout
Spar sizing
Laminate schedule
Rib spacing
EI and GJ
Weight
Margins of safety

Design Goal

Lightweight, safe, stiff, aeroelastically stable wing